SPI Design & UVM Verification - Master Interview Questions & Answers

Q1: What are the primary differences between SPI and I2C?

Answer: 1. Wires & Duplex: SPI uses 4 wires (SCLK, MOSI, MISO, CS_N) and is natively full-duplex. I2C uses 2 open-drain wires (SCL, SDA) and is half-duplex. 2. Speed: SPI easily achieves 20-50+ MHz because it uses push-pull CMOS drivers. I2C is limited by RC pullup rise times (standard: 100kbps, fast: 400kbps, high-speed: 3.4Mbps). 3. Addressing: SPI uses dedicated physical chip-select lines per peripheral (no address overhead). I2C uses in-band 7-bit or 10-bit software addressing. 4. Flow Control: I2C supports clock stretching and ACK/NACK responses. SPI has no built-in flow control or hardware acknowledgement.

Q2: Explain CPOL and CPHA and all 4 SPI Modes.

Answer: • CPOL (Clock Polarity) defines the SCLK idle voltage level: - CPOL=0: SCLK idles LOW. First edge is rising; second edge is falling. - CPOL=1: SCLK idles HIGH. First edge is falling; second edge is rising. • CPHA (Clock Phase) defines which edge samples and which edge shifts: - CPHA=0: Data is SAMPLED on the first (leading) edge and SHIFTED on the trailing edge. MSB must be valid when CS_N asserts! - CPHA=1: Data is SHIFTED on the first (leading) edge and SAMPLED on the second (trailing) edge. Modes: • Mode 0 (0,0): Idle Low, Sample Rising, Shift Falling. • Mode 1 (0,1): Idle Low, Shift Rising, Sample Falling. • Mode 2 (1,0): Idle High, Sample Falling, Shift Rising. • Mode 3 (1,1): Idle High, Shift Falling, Sample Rising.

Q3: Why is SPI called a 'Linked Shift Register' architecture?

Answer: On every serial clock cycle, one bit shifts out of the Master's shift register via MOSI into the Slave's shift register, and simultaneously one bit shifts out of the Slave's shift register via MISO into the Master. They form a circular ring buffer. A 'read' in SPI is never passive; the Master must transmit dummy bits on MOSI to generate clock pulses so the Slave can return data on MISO.

Q4: How did you verify the SPI core in UVM?

Answer: We developed a modular UVM 1.2 testbench: • UVM Agent containing Driver, Monitor, Sequencer, and Functional Coverage. • Scoreboard verifying full-duplex integrity on both MOSI and MISO paths simultaneously. • Sequences sweeping boundary vectors (0x00, 0xFF, 0xAA, 0x55), all 4 modes, and constrained-random data. • Achieved 100% test pass rate with 0 mismatches across all modes in Vivado XSim.